



2026 Biostatistics and Health Data Science Summit

Advancing Biostatistics Through Research and Collaboration

PROGRAM BOOK



14-15 September 2026

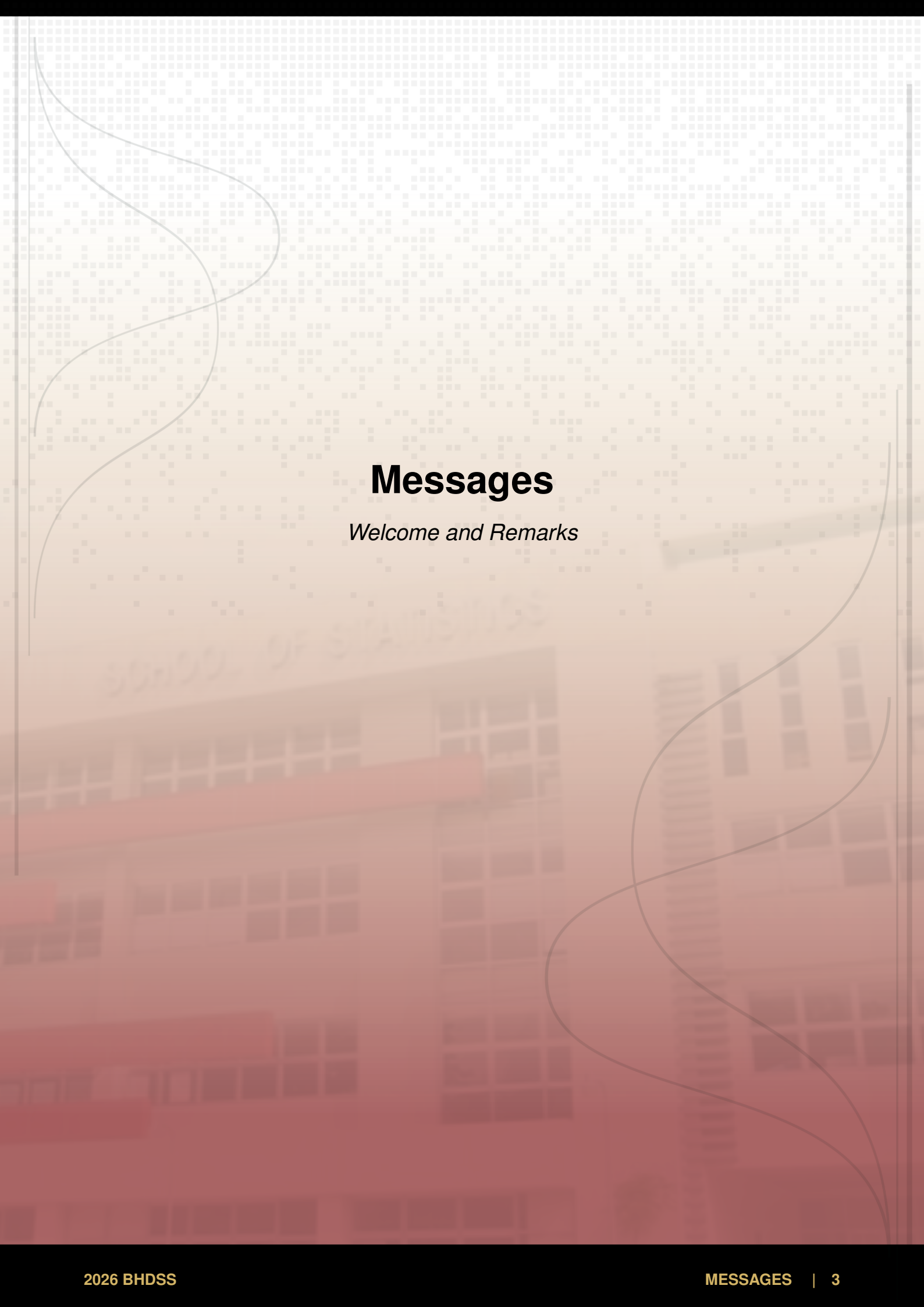


School of Statistics Auditorium
University of the Philippines Diliman
Quezon City, Philippines



biostat.upd@up.edu.ph

Messages	3
Assoc. Prof. Francisco De Los Reyes,	4
Assoc. Prof. Peter Julian Cayton, PhD	5
Assoc. Prof. Michael Daniel Lucagbo, PhD	6
Program of Activities	7
Program Overview	8
September 14, 2026 (Monday)	9
September 15, 2026 (Tuesday)	10
Research Works	11
Equivalence Testing: Bioequivalence and Beyond	12
Cost-Effectiveness Analysis: A Statistical Overview	12
Modeling Malaria Risk Near a Hydroelectric Dam: Challenges in Clustered Survival Data	13
Combining Evidence Across Heterogeneous Studies: Local Power of Exact Tests for a Common Mean	13
Advanced Spatio-Temporal Models for Misaligned Data: Applications in Real-Time Disease Surveillance	15
Simultaneous Prediction Intervals: A New Perspective on Assessing One's Health	16
Three Questions, One Disease: Methodological Lessons from Rare Disease Research	17
Multivariate Tolerance Regions for Clinical Laboratory Medicine: Regression-Based Approaches for Two-Sided and Mixed-Sided Reference Regions	18
The Data Behind the Drugstore: Mapping Pharma Performance in the Philippines	19
Bayesian Decision Limits for Correlated Analytes: A Multivariate Regression Approach with Non-Informative and Conjugate Priors	20
A Bayesian Additive Spatiotemporal Model for Young Adolescent Birth Records in the Philippines	21
Causal Mediation Analysis for Time-Varying Exposures and Confounders in Longitudinal Biostatistical Data	22
Organizing Committees	23
Sponsors	24
About the School of Statistics	25
About the Biostatistics Research Group	26



Messages

Welcome and Remarks



UNIVERSITY OF THE PHILIPPINES SCHOOL OF STATISTICS



MESSAGE

On behalf of the UP School of Statistics, it is a privilege to welcome all participants, researchers, and distinguished guests to the inaugural Biostatistics and Health Data Science Summit (BHDSS). This summit brings together scholars from local and global institutions, including UP Manila, Imperial College London, and the University of Maryland, Baltimore County. Featuring lectures by world experts Dr. Thomas Mathew, Dr. Yehe-new Kifle, and Dr. Stephen Jun Villejo, the summit showcases cutting-edge statistical methodologies applied to epidemiology and medicine. Above all, BHDSS serves as an active vehicle for research collaborations. We encourage everyone to exchange ideas, establish academic networks, and launch interdisciplinary projects that solve critical health challenges.

Beyond fostering collaborative ties, we invite you to explore the academic environment of the UP School of Statistics. Administering one undergraduate and five graduate programs, and with an increasing number of students, our institution delivers rigorous training aimed at transforming complex data into actionable public policy and solutions. This summit provides an ideal venue for prospective students to interact with our faculty, learn about ongoing healthcare research, and explore thesis mentorship opportunities. If you are considering pursuing further studies in statistics, we warmly invite you to join our master's and doctoral programs.

I would also like to acknowledge the support the UP Office of International Linkages, the UP Diliman Office of the Vice-Chancellor for Administration, the UP Statistical Center Research Foundation and the team of faculty member and staff who worked hard to make this summit possible.

Sana ay maging mabisang platform ang summit na ito para palakasin ang ating antas ng research sa biostatistics at health data science. Sa pamamagitan ng mahusay na pag-aaral, makabuluhang kolaborasyon, at tapat na paglilingkod, gamitin natin ang statistics upang magbunga ng mga makabagong solusyong mag-aangat sa kalusugan, kapakanan at dangal ng bawat Pilipino at ng buong bansa.

Assoc. Prof. FRANCISCO DE LOS REYES

Dean, UP School of Statistics



UNIVERSITY OF THE PHILIPPINES SCHOOL OF STATISTICS



MESSAGE

Warm greetings to everyone!

It is my absolute privilege to congratulate the Biostatistics Research Group of the School of Statistics for successfully bringing the inaugural Biostatistics and Health Data Science Summit 2026 (BHDSS 2026) to life. Today, students, faculty, and industry leaders gather to collaborate, exchange groundbreaking ideas, and present the latest innovations in the field of biostatistics and health data science.

The BHDSS 2026 aligns perfectly with the Research Agenda of the School of Statistics. Our institutional mission is to generate new knowledge by developing robust theories and methodologies that make complex data truly meaningful. These advancements allow us to extract actionable insights, rendering them invaluable not just within academic statistics, but as critical tools for policy and decision-making across government, enterprise, and non-governmental organizations. Through events like today's summit, our school actively champions global development goals. This summit places a sharp, necessary focus on public health and medical research.

This summit reflects the Research Group's ultimate dedication: developing statistical methods that solve real-world challenges in health and medicine. Through interdisciplinary collaboration and rigorous training, they are advancing the critical role of evidence-based, health-related decision-making.

We are deeply honored to welcome a global roster of experts to this year's summit. Our first day features the highly anticipated World Expert Lecture Series, delivered by Dr. Thomas Mathew and Dr. Yehenew Kifle of the University of Maryland, Baltimore County. The School of Statistics is ecstatic to have them, and we extend our warmest hospitality in visiting our humble institution.

The momentum continues into the second day, highlighting cutting-edge research from Dr. Stephen Jun Villejo of Imperial College London, alongside the innovative research of our own esteemed faculty. The summit serves as an excellent venue to feature the research activities done by the School specifically on biostatistics and health data science.

The Office for Research is delighted that this event both highlights state-of-the-art innovations in biostatistics and research collaborations between faculty and world experts.

Once again, congratulations to the organizers for this remarkable conference. May the BHDSS 2026 inspire lasting collaborations and fuel profound innovations!

Assoc. Prof. PETER JULIAN CAYTON, PhD

Director for Research, UP School of Statistics



UNIVERSITY OF THE PHILIPPINES SCHOOL OF STATISTICS



MESSAGE

The Biostatistics Research Group of the UP School of Statistics is delighted to host the inaugural Biostatistics and Health Data Science Summit (BHDSS). We hope that this will be the first of many editions of the summit in the years to come. The BHDSS brings together students and faculty members conducting research in biostatistics and epidemiology, as well as professionals from government and private institutions.

We are grateful to the Office of International Linkages (OIL) of the University of the Philippines for providing funding to support this initiative. We also extend our sincere appreciation to the UP School of Statistics for sponsoring the event and to the Office of the Vice Chancellor for Administration (OVCA) for the valuable support.

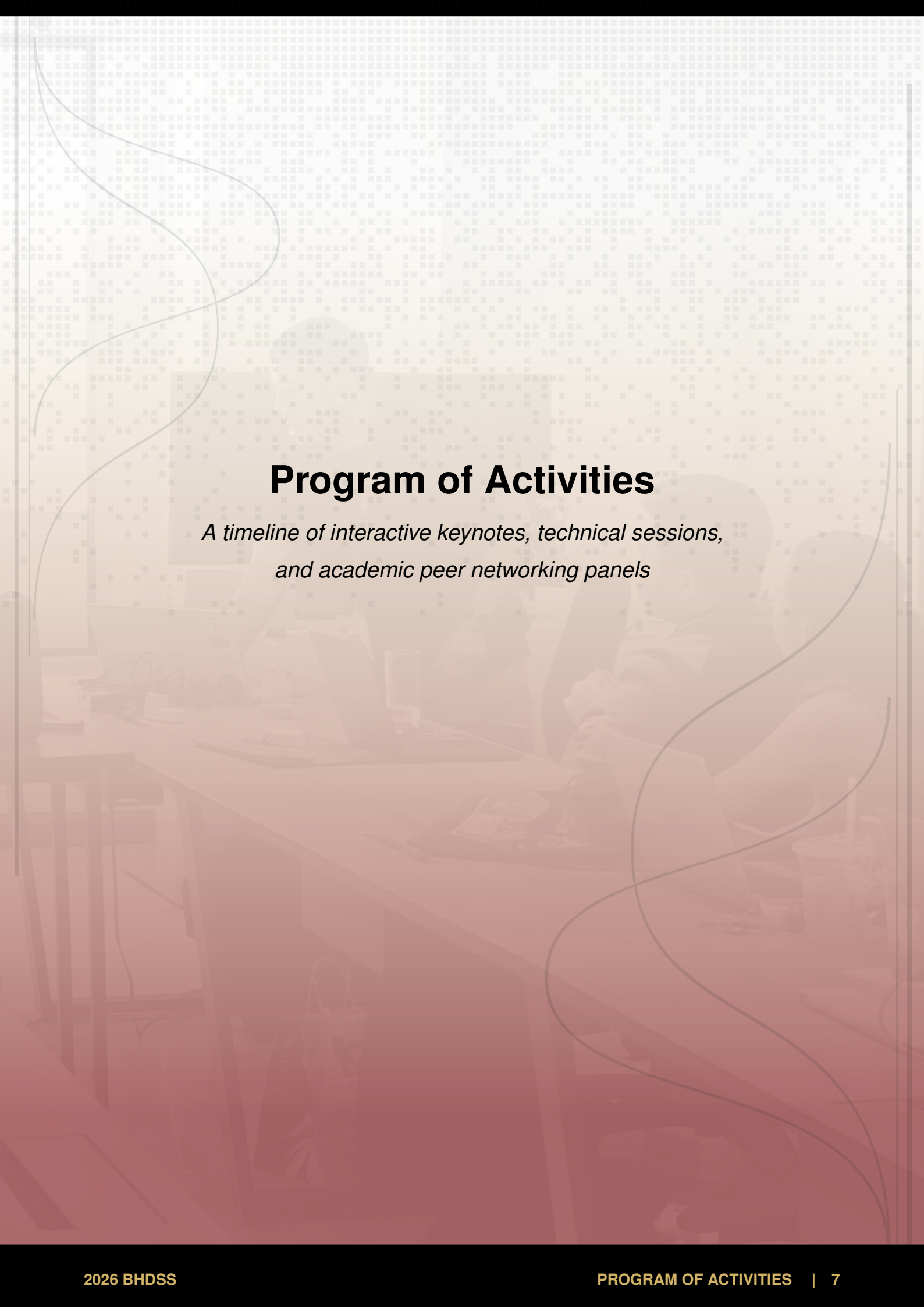
I am proud that the inaugural BHDSS features the OIL-funded World Expert Lecture Series (WELS), which includes talks by Dr. Thomas Mathew and Dr. Yehenew Kifle of the University of Maryland, Baltimore County. They will share their expertise in active areas of research in biostatistics and health data science. Both are accomplished scholars and true experts in the field. The first BHDSS also features presentations by promising early-career academics and researchers from UP Diliman, UP Manila, and Imperial College London.

Biostatistics as a discipline has grown tremendously in recent years. It plays an important role in safeguarding public health, improving patient care, and ensuring the quality and rigor of research in the biological and health sciences. Amid the growing recognition of biostatistics, however, we must not lose sight of the fact that the discipline is firmly grounded in sound statistical principles. Through this summit, we hope to showcase how rigorous statistical methodologies can be applied to address real-world problems in health and medicine.

Through the BHDSS, we hope to establish meaningful connections among researchers, students, faculty members, and professionals from different institutions. I encourage all our participants to take this opportunity to exchange ideas, explore potential research collaborations, and learn from one another. We also hope that the summit will provide opportunities for prospective graduate students to learn more about the field and connect with faculty and students from the participating institutions. Ultimately, we hope that the connections made during the BHDSS will lead not only to fruitful collaborations, but also to lasting professional relationships and friendships.

Assoc. Prof. MICHAEL DANIEL LUCAGBO, PhD

Chair, Biostatistics Research Group, UP School of Statistics
Conference Chair, Biostatistics and Health Data Science Summit



Program of Activities

*A timeline of interactive keynotes, technical sessions,
and academic peer networking panels*

Program Overview

Time	September 14, 2026 (Monday)	September 15, 2026 (Tuesday)
	Emcee: Gian Louise Roy	Emcee: Christia Lynette Piduca
7:30 am-8:30 am	 Registration & AM snacks	 Registration & AM snacks
8:30 am-9:00 am		 Dr. Stephen Jun Villejo Imperial College London
9:00 am-9:30 am	 Opening Ceremony	
9:30 am-10:30 am	 Dr. Thomas Mathew University of Maryland Baltimore County	 Dr. Michael Daniel Lucagbo University of the Philippines Diliman
10:30 am-10:45 am	 Q&A	 Q&A
10:45 am-11:15 am	 Dr. Yehenew Kifle University of Maryland Baltimore County	 Dr. Alvin Duke Sy University of the Philippines Manila
11:15 am-11:45 am		 Iana Michelle Garcia University of the Philippines Diliman
11:45 am-12:00 nn	 Q&A	 Q&A
12:00 nn-12:15 pm	 Photo Opportunity	 Lunch
12:15 pm-1:30 pm	 Lunch	
1:30 pm-2:00 pm	 Dr. Thomas Mathew University of Maryland Baltimore County	 Mark Anthony Javelosa University of the Philippines Diliman
2:00 pm-2:30 pm		 Lian Mae Tabien University of the Philippines Diliman
2:30 pm-2:45 pm	 Q&A	 Q&A
2:45 pm-3:15 pm	 Dr. Yehenew Kifle University of Maryland Baltimore County	 John Gilbert Rabanal University of the Philippines Diliman
3:15 pm-3:45 pm		 John Eustaquio University of the Philippines Diliman
3:45 pm-4:00 pm	 Q&A	 Q&A
4:00 pm-6:00 pm	 Socials & Networking	 Closing Ceremony
6:00 pm	 Conference Dinner For invited speakers, organizers, & UP Stat faculty	

September 14, 2026 (Monday)

Emcee: Gian Louisse Roy

7:30 am-9:00 am



Registration & AM snacks

Distribution of conference materials, badges, and credentials at the main lobby

9:00 am-9:30 am



Opening Ceremony

National anthem and invocation; Welcome remarks; Introduction of speakers

9:30 am-10:30 am



Research Talk

Equivalence Testing: Bioequivalence and Beyond by Dr. Thomas Mathew

10:30 am-10:45 am



Q&A

10:45 am-11:45 am



Research Talk

Modeling Malaria Risk Near a Hydroelectric Dam: Challenges in Clustered Survival Data by Dr. Yehenew Kifle

11:45 am-12:00 nn



Q&A

12:00 nn-12:15 pm



Photo Opportunity

12:15 pm-1:30 pm



Lunch

1:30 pm-2:30 pm



Research Talk

Cost-Effectiveness Analysis: A Statistical Overview by Dr. Thomas Mathew

2:30 pm-2:45 pm



Q&A

2:45 pm-3:45 pm



Research Talk

Combining Evidence Across Heterogeneous Studies: Local Power of Exact Tests for a Common Mean by Dr. Yehenew Kifle

3:45 pm-4:00 pm



Q&A

4:00 pm-6:00 pm



Socials & Networking

Time to engage with others at the lobby!

6:00 pm



Conference Dinner

For invited speakers, organizers, and UP Stat faculty

September 15, 2026 (Tuesday)

Emcee: Christia Lynette Piduca

7:30 am-8:30 am



Registration & AM snacks

Distribution of conference materials, badges, and credentials at the main lobby

8:30 am-9:30 am



Research Talk

Advanced Spatio-Temporal Models for Misaligned Data: Applications in Real-Time Disease Surveillance by Dr. Stephen Jun Villejo

9:30 am-10:30 am



Research Talk

Simultaneous Prediction Intervals: A New Perspective on Assessing One's Health by Dr. Michael Daniel Lucagbo

10:30 am-10:45 am



Q&A

10:45 am-11:15 am



Research Talk

Three Questions, One Disease: Methodological Lessons from Rare Disease Research by Dr. Alvin Duke Sy

11:15 am-11:45 am



Research Talk

Multivariate Tolerance Regions for Clinical Laboratory Medicine: Regression-Based Approaches for Two-Sided and Mixed-Sided Reference Regions by Iana Michelle Garcia

11:45 am-12:00 nn



Q&A

12:00 nn-1:30 pm



Lunch

1:30 pm-2:00 pm



Research Talk

The Data Behind the Drugstore: Mapping Pharma Performance in the Philippines by Mark Anthony Javelosa

2:00 pm-2:30 pm



Research Talk

Bayesian Decision Limits for Correlated Analytes: A Multivariate Regression Approach with Non-Informative and Conjugate Priors by Lian Mae Tabien

2:30 pm-2:45 pm



Q&A

2:45 pm-3:15 pm



Research Talk

A Bayesian Additive Spatiotemporal Model for Young Adolescent Birth Records in the Philippines by John Gilbert Rabanal

3:15 pm-3:45 pm



Research Talk

Causal Mediation Analysis for Time-Varying Exposures and Confounders in Longitudinal Biostatistical Data by John Eustaquio

3:45 pm-4:00 pm



Q&A

4:00 pm-6:00 pm



Closing Ceremony

Awarding of certificates; Closing message

Research Works

*Showcasing academic studies driving meaningful innovation
in biostatistics and its applications*



Dr. Thomas Mathew

Professor

University of Maryland Baltimore County

🕒 September 14 | 9:30 am-10:30 am

Equivalence Testing: Bioequivalence and Beyond

In hypothesis testing problems, the null hypothesis is often a statement of equality. The two-sample t-test is a classic example where the null hypothesis states that two population means are equal. In some applications on the comparison of two means, it is more appropriate to test if the two means are equivalent; i.e., if they are close according to a specified threshold. Many applications call for testing equivalence, rather than equality, where the statement of equivalence is the alternative hypothesis so that equivalence is concluded by rejecting the null hypothesis. In the talk I will give a few applications where equivalence testing is required. These will include testing bioequivalence, testing the equivalence of several normal means relevant to the comparison of several treatments in a clinical trial, testing the equivalence of covariance matrices, comparing drug dissolution profiles, and equivalence testing for food safety assessment. All of the problems will be motivated and illustrated using practical applications.

🕒 September 14 | 1:30 pm-2:30 pm

Cost-Effectiveness Analysis: A Statistical Overview

Identifying treatments or interventions that are cost-effective (more effective at a reasonable cost) is clearly important in health policy decision making, especially in the allocation of health care resources. Various measures of cost-effectiveness that are informative, intuitive and simple to explain have been suggested in the literature. Popular and widely used measures include the incremental cost-effectiveness ratio (ICER), defined as the ratio between the difference of average costs and the difference of average effectiveness in two populations receiving two treatments. The ICER is interpreted as the additional cost per unit of effectiveness gained. Yet another measure is the incremental net benefit (INB), which is the difference between the incremental cost and the incremental effectiveness after multiplying the latter with a “willingness-to-pay” amount. In the talk, I will provide a selected review of the statistical criteria and methodologies for cost-effectiveness analysis. In particular, some of the recently introduced probabilistic criteria will be discussed and examples will be given.

About the speaker

Thomas Mathew is Professor, Department of Mathematics & Statistics, University of Maryland Baltimore County. He earned his PhD in statistics from the Indian Statistical Institute in 1983, and has been a faculty member at the University of Maryland since 1985. His research interests are on both methodological and applied topics, including cost-effectiveness analysis, equivalence testing, exposure data analysis, meta-analysis, mixed and random effects models, and tolerance intervals. He is the co-author of two books: *Statistical Tests in Mixed Linear Models* and *Statistical Tolerance Regions: Theory, Applications and Computation*, both published by Wiley. He has served on the Editorial Boards of several journals, and is currently an Associate Editor of the *Journal of Multivariate Analysis* and *Sankhya*. Dr. Mathew is a Fellow of the American Statistical Association, and a Fellow of the Institute of Mathematical Statistics. He is a two-time winner of the American Statistical Association's Youden Award in Interlaboratory Testing. He has also been appointed as Presidential Research Professor at his campus.



Dr. Yehenew Kifle

*Associate Professor
University of Maryland Baltimore County*

🕒 September 14 | 10:45 am-11:45 am

Modeling Malaria Risk Near a Hydroelectric Dam: Challenges in Clustered Survival Data

Understanding how large hydroelectric dams influence malaria risk is important for public health planning in malaria-endemic regions. This study investigates the association between household proximity to the Gilgel Gibe hydroelectric dam in Ethiopia and time to malaria infection. The analysis is complicated by clustering of individuals within villages and confounding between village membership and distance from the dam.

We compare marginal, fixed-effects, stratified, and frailty survival models for analyzing the clustered time-to-event data. The models produce notably different estimates of the effect of distance from the dam. In particular, the standard frailty model estimates a weighted combination of within- and between-village distance effects, which can be misleading when these effects differ. A between-within frailty formulation provides a useful approach for separating these sources of variation. The findings highlight the importance of appropriately accounting for clustering and covariate structure when analyzing spatially clustered infectious-disease survival data.

🕒 September 14 | 2:45 pm-3:45 pm

Combining Evidence Across Heterogeneous Studies: Local Power of Exact Tests for a Common Mean

Combining evidence across independent studies is a fundamental problem in biostatistics, particularly when studies share a common underlying effect but exhibit heterogeneous and unknown variances. Such settings arise naturally in meta-analysis, multi-center studies, laboratory investigations, and the integration of health data from multiple populations or data sources. This work investigates exact inference for a common mean across independent normal populations with unequal variances and compares the local power properties of six exact testing procedures.

We consider several approaches for combining study-specific evidence, including Tippett's and Wilkin-son's procedures, Fisher's method, the inverse normal method, and modified tests based on weighted combinations of component test statistics. Analytical expressions for the local power of the six proce-dures are derived under alternatives close to the null hypothesis. For equal sample sizes, the resulting expressions permit a uniform comparison of the tests irrespective of the unknown population variances. Numerical evaluations further examine their performance under both equal and unequal sample-size settings.

The results show that the modified exact tests consistently provide the strongest local power among the procedures considered, while the inverse normal method also performs competitively and offers computational simplicity. These findings provide practical guidance for combining evidence across

heterogeneous studies and highlight the importance of selecting statistically efficient procedures when integrating information from multiple biomedical and health data sources.

About the speaker

Yehenew G. Kifle is a tenured Associate Professor in the Department of Mathematics & Statistics at the University of Maryland, Baltimore County, USA. He earned his PhD in Statistical Data Analysis from Ghent University, Belgium, in 2013, and joined UMBC in 2018. Prior to joining UMBC, he held faculty positions at the University of Limpopo in South Africa and Jimma University in Ethiopia.

His research interests include survival and frailty models, longitudinal and correlated data analysis, Bayesian and mixed-effects modeling, statistical meta-analysis and machine learning methods, with applications in public health, infectious diseases, environmental sciences, and agriculture. He has published extensively in statistical, biomedical, and interdisciplinary journals and has supervised several PhD and master's students in the United States and Africa.

Dr. Kifle has led research and capacity-building projects funded by the American Statistical Association (ASA), Institute of Mathematical Statistics (IMS), Carnegie Corporation of New York, and other organizations. He is Chair and overall organizer of the African International Conference in Statistics (AIC), which promotes statistical research, education, international collaboration, and capacity building in Africa.



Dr. Stephen Jun Villejo

*Research Associate
Imperial College London*

🕒 September 15 | 8:30 am-9:30 am

Advanced Spatio-Temporal Models for Misaligned Data: Applications in Real-Time Disease Surveillance

This talk comprises two major parts, both centred around the overarching theme of real-time disease surveillance. The key statistical challenge I address is the spatial and temporal misalignment of the data.

The first part considers the problem of estimating a latent spatially continuous process from aggregated outcome data. I adopt a block aggregation framework that separates the latent process model from the sampling model, and estimate it using a linearised version of Integrated Nested Laplace Approximation (INLA). I illustrate this approach by predicting SARS-CoV-2 viral load, measured as the number of gene copies, over arbitrary areal configurations in the United Kingdom using aggregated data from the catchment areas of a sparse network of sewage treatment works. To address the difficulty arising from the fact that log-Gaussian outcomes are not closed under aggregation, I use moment matching to estimate the sampling model parameters and propose an iterative procedure to adjust the non-linear predictor expression and the scaling of the likelihood precision. This modelling approach provides a coherent framework for inference, prediction over new areal units, and spatial disaggregation.

The second part of my talk focuses on temporal misalignment in lagged covariate-response relationships, particularly in settings where the response variable is observed at a lower frequency than the predictors. In disease surveillance, for example, clinical outcomes may be recorded weekly and forecast at a one-week horizon, whereas environmental indicators such as pollution levels and wastewater viral loads are available daily. To address this, I extend the Mixed-Data Sampling framework to a spatial setting through a Spatial Distributed-Lag Mixed-Data Sampling (SDL-MIDAS) model. I discuss an efficient Bayesian computational tool for fitting these models, together with an accompanying R package that I am currently developing. An application from Wales shows that SDL-MIDAS models improve forecasts of weekly hospital admissions and positivity rates using daily wastewater signals as predictors.

Overall, this talk discusses flexible and principled approaches for handling spatial and temporal misalignment, with important applications in integrating environmental surveillance data into public health monitoring systems.

About the speaker

Stephen Jun Villejo is a Postdoctoral Research Associate in Biostatistics at Imperial College London and a member of the Environment and Health Statistics Group within the Medical Research Council Centre for Environment & Health. His current research, funded by UK Research and Innovation, focuses on advancing spatio-temporal methods motivated by wastewater-based epidemiology. He works with collaborators from the UK Health Security Agency and Public Health Wales to integrate these models into a real-time public health surveillance system. He completed his PhD in Statistics at the University of Glasgow and previously served as an Assistant Professor at the UP School of Statistics. His research interests include methods for spatially and temporally misaligned data, data fusion models, and uncertainty propagation in multi-stage Bayesian models.



Dr. Michael Daniel Lucagbo

*Associate Professor
University of the Philippines Diliman*

🕒 September 15 | 9:30 am-10:30 am

Simultaneous Prediction Intervals: A New Perspective on Assessing One's Health

Reference intervals are often considered the most widely used medical decision-making tool. They are used to assess the health status of individuals based on laboratory measurements. Individuals whose analyte values fall inside the reference interval are diagnosed as being healthy. On the other hand, those whose values fall outside the interval may be considered as having abnormal health status. Reference intervals are computed so that they contain the central 95% of the healthy population or reference population. Therefore, the endpoints of the reference interval are the 2.5th and 97.5th population percentiles. Since these percentiles are unknown in actual practice, the reference interval is estimated from a random sample. When using estimated percentiles, the resulting reference interval will cover less than 95% of the population. To address this problem of liberal coverage, a recommended solution is to use a 95% prediction interval.

When multiple analytes are needed to diagnose a complex condition accurately, simultaneous prediction intervals whose joint coverage is 95% are needed. In this presentation, we consider several approaches for constructing simultaneous prediction intervals under a broad range of settings, including the multivariate normal and nonparametric settings. The methods to be presented accommodate correlation among analytes, which is particularly relevant when multiple biomarkers are used to assess a complex medical condition. Furthermore, we extend the methodologies to the case where covariate information is available. Finally, we evaluate the methodologies using Monte Carlo simulations that report the estimated coverage probabilities.

About the speaker

Michael Daniel Lucagbo is currently an Associate Professor at the School of Statistics, University of the Philippines, where he is also the head of the Biostatistics Research Group. He received his PhD in Statistics from the University of Maryland, Baltimore County (UMBC). He is also a graduate of the BS Statistics and MS Statistics programs of the UP School of Statistics. He has worked as research fellow for the Office of Biostatistics, U.S. Food and Drug Administration. He is a recipient of the American Statistical Association (ASA) Biopharmaceutical Section Scholarship Award. His research areas include multivariate analysis and biostatistics.



Dr. Alvin Duke Sy

***Assistant Professor
University of the Philippines Manila***

🕒 September 15 | 10:45 am-11:15 am

Three Questions, One Disease: Methodological Lessons from Rare Disease Research

Rare disease research presents methodological challenges that are seldom illustrated in methodological textbooks. Using a single-center Philippine gestational trophoblastic disease registry as a case study, this lecture reflects on a series of analytical decisions made while investigating time to remission under the constraints of limited data. Rather than comparing statistical methods exhaustively, the presentation highlights how different approaches addressed different questions encountered during the course of the research, and discusses the methodological considerations that informed those choices. The lecture is intended for researchers interested in the practical application of statistical methods in rare disease and small-sample clinical research.

About the speaker

Dr. Alvin Duke R. Sy is a nurse-biostatistician and Assistant Professor at the Department of Epidemiology and Biostatistics, College of Public Health, University of the Philippines Manila.

He earned his Bachelor of Science in Nursing from the UP College of Nursing, placing eighth in the 2011 Philippine Nurse Licensure Examination, and has worked as a staff nurse at the Philippine General Hospital before pursuing graduate training in biostatistics and clinical research. He holds a Master of Science in Public Health (Biostatistics) and a Doctor of Philosophy in Health Sciences by Research, with a concentration in Clinical Research, from the University of the Philippines Manila.

His research focuses on statistical methods for clinical and public health research, particularly survival analysis, prognostic modeling, biomarker evaluation, and maternal and reproductive health. He also teaches courses in biostatistics, epidemiology, and research methods, and serves as a statistical consultant for clinical and public health research.



Iana Michelle Garcia

*Assistant Professor
University of the Philippines Diliman*

🕒 September 15 | 11:15 am-11:45 am

Multivariate Tolerance Regions for Clinical Laboratory Medicine: Regression-Based Approaches for Two-Sided and Mixed-Sided Reference Regions

Co-author: Michael Daniel Lucagbo

In clinical laboratory medicine, analyte measurements are commonly evaluated using univariate reference intervals. However, when multiple analytes are required for diagnosis, applying separate univariate reference intervals increases the occurrence of false-positive results. Multivariate reference regions (MRRs) are more appropriate in this context because they account for the cross-correlations among analytes. Compared with conventional ellipsoidal MRRs, rectangular MRRs are easier to interpret and facilitate the identification of outlying analyte measurements. Prediction- and tolerance-based approaches have been proposed for constructing rectangular MRRs, with tolerance regions possessing a multiple-use property that makes them suitable for repeated application across multiple patients. In addition, analyte reference limits may be two-sided or one-sided depending on the biomarker and disease context. This paper focuses on regression-based rectangular MRRs constructed using multivariate tolerance criteria while accounting for covariate information. Specifically, we present regression-based multivariate tolerance regions, multivariate central tolerance regions, and mixed-sided multivariate tolerance regions. The performance of these proposed approaches is evaluated through simulation studies, and results show that all methods achieve coverage probabilities close to the nominal level. Finally, we apply the proposed approaches to a clinical laboratory dataset.

About the speaker

Iana Michelle Garcia is an Assistant Professor at the University of the Philippines School of Statistics. She earned her Master of Science in Statistics from the same institution and was a recipient of the Accelerated Science and Technology Human Resource Development Program (ASTHRDP) scholarship from the Department of Science and Technology Science Education Institute (DOST-SEI). She is a member of the Pi Gamma Mu International Honor Society.

Her research interests include biostatistics, computational statistics, tolerance regions, and statistical machine learning. She has co-authored a published work on tolerance regions and continues to explore their applications in both theoretical and applied contexts. She has presented her research at the International Conference on Advanced Machine Learning and Data Science (AMLDS) in Tokyo, Japan; the 16th National Convention on Statistics (NCS) organized by the Philippine Statistics Authority (PSA); and the 33rd International Biometric Conference in Seoul, South Korea.

In her teaching and extension activities, she has handled courses in programming, computational statistics, and statistical theory, and has served as a resource person for R programming training programs.



Mark Anthony Javelosa

*Assistant Professor
University of the Philippines Diliman*

🕒 September 15 | 1:30 pm-2:00 pm

The Data Behind the Drugstore: Mapping Pharma Performance in the Philippines

Every time a patient buys medicine at a local drugstore, a complex chain of data is set into motion. For pharmaceutical companies operating in the Philippines, tracking whether their products are succeeding requires navigating a massive, interconnected puzzle of information. In this talk, we shall be introduced to the fascinating world of commercial market intelligence and the diverse data ecosystems that power a multi-billion-peso industry.

About the speaker

Mark Anthony Javelosa earned both his Bachelor of Science and Master of Science in Statistics from the University of the Philippines Diliman.

He served as a senior lecturer of statistics at the Department of Clinical Epidemiology of the College of Medicine in the University of the Philippines Manila, at the Ateneo Graduate School of Business - Center for Continuing Education, and at La Consolacion College Bacolod City. He previously held key leadership and technical roles in the pharmaceutical industry, including Senior Biostatistician at Pharmaceutical Product Development (PPD), Consultant for Johnson & Johnson Philippines, and most recently Data Analytics Manager at iNovaPharmaceuticals Philippines, Inc.

His professional interests include biostatistics, clinical epidemiology and healthcare research. He also co-authored several peer-reviewed publications in primary care, public health, and clinical research, contributing to evidence-based healthcare in the Philippines.



Lian Mae Tabien

*Assistant Professor
University of the Philippines Diliman*

🕒 September 15 | 2:00 pm-2:30 pm

Bayesian Decision Limits for Correlated Analytes: A Multivariate Regression Approach with Non-Informative and Conjugate Priors

Co-author: Michael Daniel Lucagbo

Decision limits play a crucial role in laboratory medicine by guiding diagnostic and clinical decision-making processes. Unlike reference intervals, decision limits are often one-sided thresholds, typically expressed as prediction or tolerance intervals. However, in complex diagnoses requiring multiple correlated analyte measurements, constructing separate univariate intervals may increase false positives. It is recommended to construct multivariate statistical regions to account for the cross-correlations among analytes. Hence, this research proposes a Bayesian multivariate regression framework for constructing combined statistical decision limits that incorporates covariates (e.g., age and sex). The methodology operates under a multivariate normal model and allows for prior specification using both non-informative and conjugate priors. Results from simulation studies indicate that limits derived under non-informative priors have highly satisfactory frequentist properties and outperform benchmark methods even under weak cross-correlations. In contrast, limits based on conjugate priors tend to underperform in small samples due to prior influence, although this effect diminishes with larger sample sizes. The proposed method is also computationally efficient, producing decision limits in under a minute, highlighting its practical feasibility for clinical application. Overall, this Bayesian multivariate approach provides a principled and adaptable framework for generating accurate decision limits in complex diagnostic contexts.

About the speaker

Lian Mae T. Tabien is an Assistant Professor at the University of the Philippines Diliman - School of Statistics. She graduated Summa Cum Laude with a Bachelor of Science in Statistics in 2019 and earned her Master of Science in Statistics in 2024 in the same university. She teaches foundational courses in statistics, primarily for undergraduate students in the school. Her research interests include biostatistics, with a focus on laboratory medicine, and multivariate analysis.



John Gilbert Rabanal

*Assistant Professor
University of the Philippines Diliman*

🕒 September 15 | 2:45 pm-3:15 pm

A Bayesian Additive Spatiotemporal Model for Young Adolescent Birth Records in the Philippines

Co-author: Welfredo Patungan

Quantitative studies about young adolescent pregnancy in the Philippines had been done through the analysis of survey data. Official birth records provide a more advantageous data source in examining the dynamics of the young adolescent pregnancy due to regular recording of births. In this study, we propose a statistical methodology utilizing official birth records through a Bayesian hierarchical framework. The method accounts for the unique features of the aggregated young adolescent birth counts such as being a spatiotemporal data with known upper bounds and being characterized by the presence of excess zero counts. Our framework consists of the binomial hurdle model and Poisson hurdle model with offset, the Leroux model for spatial effects (Leroux et al., 2000), and an additive nonparametric trend model for temporal effects (Knorr-Held, 2000). We analyze and assess the performance of the proposed models using the quarterly data on young adolescent birth counts in Luzon, Philippines from 2006 to 2019 via Hamiltonian Monte Carlo. Areas that are more prone to observe a young adolescent birth and a temporal pattern about the occurrence of young adolescent births are identified.

About the speaker

John Gilbert Rabanal is currently serving as assistant professor and deputy director for facilities and resource management at the UP School of Statistics. He earned his Bachelor of Science in Statistics degree in 2019, graduating Cum Laude. He also obtained his Master of Science in Statistics degree in January 2025 from the same institution. He teaches a diverse range of courses, from elementary statistics and probability theory to matrix theory and Bayesian inference. His research interests include Bayesian statistics, spatiotemporal analysis, categorical data analysis, and applications of statistical methods in the social sciences. He has actively contributed to the academic community through presentations at both local and international conferences.



John Eustaquio

*Assistant Professor
University of the Philippines Diliman*

🕒 September 15 | 3:15 pm-3:45 pm

Causal Mediation Analysis for Time-Varying Exposures and Confounders in Longitudinal Biostatistical Data

Understanding the causal mechanisms behind biomedical research often requires detaching how exposures exert their effects on health outcomes over time through intermediate processes. However, in longitudinal studies, both exposures and confounders frequently vary with time which creates a complex feedback structures that would make standard mediation approaches invalid. This study developed a novel framework for causal mediation analysis in longitudinal biostatistical data with time-varying exposures and confounders by integrating marginal structural models (MSM) with structural nested models (SNM), and targeted maximum likelihood estimation (TMLE) in a single causal inference paradigm. A flexible estimation strategy was proposed that utilizes machine learning-based nuisance estimation to improve the robustness and efficiency but still maintaining valid causal interpretation under realistic longitudinal data structures. Simulation studies demonstrated that the proposed method provides unbiased estimates of direct and indirect effects even under complex time-dependent confounders. The methodology is applied to simulated data mimicking a large cohort study of HIV treatment dynamics, where the causal pathways linking antiretroviral therapy adherence, immune response, and virologic suppression are examined. The results shown substantial time-varying mediation effects which highlight the importance of accounting for temporal effect in biomedical causal inference.

About the speaker

John De Guzman Eustaquio is an Assistant Professor at the University of the Philippines School of Statistics. He holds master's degrees in Statistics from the University of Maryland, Baltimore County and the University of the Philippines Diliman where he is currently pursuing his PhD in Data Science. His research interests include survival analysis, meta-analysis, biostatistics, computational statistics, labor and employment statistics, and customer satisfaction and behavior. He has published research on clustered survival models, stochastic frontier analysis, demographic behavior, regional development, and machine learning applications in health research. In addition to his academic work, he has served as a statistician, survey methodologist, and Computer-Assisted Personal Interviewing instrument developer for numerous government agencies and national research projects. He is a member of several professional organizations, including the Philippine Statistical Association, International Statistical Institute, International Association for Statistical Computing, and International Biometric Society.

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About the School of Statistics



Mission

The School of Statistics aims to advance the development and maintenance of a high level of efficiency in statistical activities in the country and in the Asia-Pacific region through its academic programs as well as research and extension services.

Vision

We envision a School that plays a dominant role in the creation of a Filipino society that is literate in statistics and that actively participates in the development of statistics in the Asia-Pacific region.

History

In December 1953, the Statistical Training Center was established under a bilateral agreement between the Philippine Government and the United Nations. It has since grown into a degree-granting institution and in 1998 the Center was renamed as School of Statistics. The School's mission is to advance the development and maintenance of a high level of efficiency in statistical activities in the country and in the Asia-Pacific Region through its academic programs, research, and extension services.

The School of Statistics is recognized by Commission on Higher Education as a Center of Excellence in Statistics. In 2014, the B.S. Statistics program received international accreditation from the ASEAN Universities Network. The School offers one undergraduate program and five graduate programs. These are Bachelor of Science (Statistics) program, the Master of Science (MOS) program, the Master of Science (Statistics) program, the Doctor of Philosophy (Statistics) program, the Professional Master in Data Science (Analytics), and the Doctor of Philosophy (Data Science).

Employing top calibre faculty members, and with internationally recognized programs, it has produced numerous distinguished and respected statisticians in various profession and is widely recognized as the primary source of statistical expertise for the Philippine Statistical System and Industry.



UPDStat

statinfo.upd@up.edu.ph

8981-8500 local 3503/3504

(632) 8928-0881

About the Biostatistics Research Group



The Biostatistics Research Group of the UP School of Statistics focuses on the development and application of statistical methods to address challenges in health, medicine, and public health research. Through methodological innovation, interdisciplinary collaboration, and research training, the group advances the role of biostatistics in generating evidence and informing health-related decisions.

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